

YANBU PRINCE ABDULAZIZ, Saudi Arabia

WMO: 404390

Lat: 24.144N

Lon: 38.063E

Elev: 8

StdP: 101.23

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	11.9	13.1	-3.2	2.9	25.3	-0.8	3.5	24.6	11.4	23.9	10.4	23.3	1.3	30	0.465

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.3	44.4	24.2	43.0	24.3	41.8	24.4	30.5	36.0	29.7	35.3	29.0	34.6	7.8	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
29.2	25.9	33.4	28.2	24.5	32.7	27.8	23.9	32.4	102.9	35.4	98.7	35.1	94.9	35.0	34.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
12.0	10.6	9.5	9.2	47.5	1.4	1.5	8.2	48.5	7.4	49.4	6.6	50.2	5.6	51.3		
			5.3	31.0	1.6	1.4	4.2	32.0	3.3	32.8	2.4	33.6	1.3	34.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	28.6	21.3	22.5	24.6	27.8	31.4	33.3	33.9	34.5	33.4	30.6	26.5	23.1
	DBStd	5.24	2.54	2.59	2.74	2.68	2.55	2.37	2.03	2.28	2.00	1.96	2.31	2.38
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	5	4	1	0	0	0	0	0	0	0	0	0	1
	CDD10.0	6788	349	349	452	535	663	700	739	760	702	637	494	406
	CDD18.3	3752	95	117	194	285	405	450	481	502	452	379	244	148
	CDH23.3	52217	819	1081	2096	3615	5918	7029	7622	8062	6870	5143	2632	1330
Wind	WSAvg	3.8	3.1	3.5	3.8	3.8	4.1	4.7	4.6	4.6	4.1	3.3	3.1	2.9
	PrecAvg	33	8	1	5	1	1	0	0	0	0	2	7	7
	PrecMax	124	80	11	93	9	19	0	0	2	1	25	48	51
	PrecMin	2	0	0	0	0	0	0	0	0	0	0	0	0
	PrecStd	33	16	3	18	2	4	0	0	0	0	5	12	14
	DB	32.0	33.9	37.0	39.2	44.0	45.9	45.9	46.9	44.8	41.8	37.8	34.0	
	MCWB	18.4	18.1	19.5	21.1	22.3	23.0	23.9	25.0	24.3	23.5	20.1	18.9	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	30.2	31.8	34.5	37.2	41.9	43.9	43.2	44.1	42.8	39.2	35.4	32.1	
	2% MCWB	18.8	18.5	19.7	21.7	22.5	23.5	24.7	25.5	24.7	22.7	21.0	19.3	
	5% DB	29.0	30.2	33.0	36.0	39.9	42.1	41.8	42.1	41.1	37.9	34.0	30.9	
	5% MCWB	18.9	19.1	20.1	21.7	23.0	23.8	24.7	25.5	24.8	23.9	21.4	19.6	
	10% DB	27.8	29.0	31.2	34.6	38.1	40.4	40.0	40.2	39.2	36.4	32.8	29.2	
	10% MCWB	18.4	18.9	19.6	21.3	22.6	23.7	24.5	25.3	25.1	24.5	21.6	19.1	
	WB	24.1	24.2	25.0	26.6	28.5	29.6	30.7	31.5	31.1	30.6	27.6	25.1	
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	MCDB	27.8	28.7	30.0	32.3	34.6	36.6	36.7	37.2	36.9	35.5	32.2	29.1	
	2% WB	22.5	22.8	23.7	25.2	27.2	28.4	29.7	30.6	29.8	29.0	25.8	23.3	
	2% MCDB	26.9	27.6	29.6	31.6	33.7	35.3	35.6	36.0	35.2	33.8	30.9	28.2	
	5% WB	21.1	21.8	22.7	24.2	26.1	27.5	28.8	29.9	29.0	27.8	24.6	22.1	
	5% MCDB	26.4	27.1	29.2	31.2	33.2	34.2	34.5	35.3	34.4	32.7	30.0	27.5	
	10% WB	19.7	20.7	21.5	23.3	25.3	26.7	27.9	29.1	28.1	26.9	23.6	20.8	
	10% MCDB	25.6	26.6	28.5	30.9	32.8	33.8	33.6	34.8	33.8	32.0	29.3	26.8	
Mean Daily Temperature Range	MDBR	12.7	13.0	13.6	13.5	13.8	14.4	12.9	12.3	13.0	12.7	12.9	13.1	
	5% DB	13.3	13.9	14.6	14.5	15.5	16.3	15.3	14.6	15.0	14.5	14.1	14.4	
	MCWBR	7.3	7.3	7.2	6.9	7.2	7.8	7.7	7.7	7.5	7.8	6.9	7.2	
	5% WB	10.7	10.6	12.1	12.4	13.5	14.3	12.7	10.9	11.9	12.0	12.0	11.4	
	MCWBR	7.2	7.0	7.4	7.3	7.2	7.5	7.2	6.9	6.8	7.2	6.5	6.6	
Clear-Sky Solar Irradiance	taub	0.390	0.449	0.480	0.517	0.534	0.522	0.550	0.592	0.542	0.500	0.435	0.410	
	taud	2.280	2.077	1.980	1.915	1.877	1.916	1.888	1.807	1.888	1.991	2.162	2.210	
	Ebn at Noon	861	830	823	800	782	785	764	732	759	770	804	820	
	Edh at Noon	120	157	181	197	204	195	200	217	196	168	134	124	
All-Sky Solar Radiation	RadAvg	4.52	5.40	6.55	7.29	7.69	8.04	7.73	7.20	6.64	5.72	4.75	4.25	
	RadStd	0.23	0.21	0.25	0.23	0.25	0.12	0.15	0.17	0.17	0.14	0.15	0.11	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.50	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+168	+176
(47)	Regional (1 neighbor)	+0.46	N/A	-0.86	N/A	N/A	N/A	N/A	N/A	+156	+153

Nomenclature: See separate page